

Bingnan Li

Ph.D. Student in Cognitive Science, University of California, San Diego

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RESEARCH INTERESTS

Generative models and controllable image & video generation, including diffusion distillation, layout-to-image generation, and multimodal generative systems.

EDUCATION

University of California, San Diego

Ph.D. in Cognitive Science · Advisor: Prof. Zhuowen Tu

Sep. 2026 – Present

San Diego, CA

University of California, San Diego

M.S. in Computer Science and Engineering · GPA: 3.91/4.00

Sep. 2024 – Jun. 2026

San Diego, CA

ShanghaiTech University

B.E. in Computer Science and Technology · GPA: 3.87/4.00 · Rank: 3/248

Sep. 2020 – Jul. 2024

Shanghai, China

RESEARCH EXPERIENCE

Qwen Application, Alibaba Group

Research Intern · Mentor: Jiaming Liu

Summer 2026

Shanghai, China

- Led a study of classifier-free guidance (CFG) in on-policy diffusion distillation (OPD), showing that matching CFG-composed teacher and student velocities is under-identified: positive- and negative-branch errors can cancel each other out.
- Identified **Negative Branch Asymmetry (NBA)**, a failure mode that arises when the teacher's negative branch carries privileged information unavailable to the student, driving antagonistic branch-error dynamics.
- Proposed **Positive-Direction Matching (PDM)**, a branch-aware OPD objective that separately constrains the positive prediction and the CFG conditional direction, enabling robust distillation for dense-to-sparse video control across guidance scales [3].
- Contributed to agent-native video representation learning [1] and visually aligned follow-up suggestions for conversational image editing [2].

Amazon Web Services, AWS AI Labs

Applied Scientist Intern · Mentors: Zhaoyang Zhang, Yantao Shen, Zhuowen Tu

Jul. 2025 – Dec. 2025

Seattle, WA

- Built a multi-expert LLM ensemble that fuses domain-specialized experts at the logit level, with efficient parallel inference.
- Quantified oracle-ensemble upper bounds to measure the attainable gain beyond any single expert.
- Explored dynamic ensembling strategies, including logit-level weight prediction, trajectory-level quality estimation, and rollout-based correctness scoring.
- Showed that dynamic ensembles outperform uniform ensembles and show clear potential to surpass the best individual expert on reasoning tasks [5].

Machine Learning, Perception, and Cognition Lab (MLPC), UC San Diego

Graduate Researcher · Advisor: Prof. Zhuowen Tu

2024 – Present

San Diego, CA

- Investigated failure modes of layout-to-image models under dense, overlapping bounding boxes and proposed **OverLayScore**, a spatial-semantic metric of overlap difficulty.
- Built **OverLayBench**, a difficulty-balanced benchmark with dense captions, relationship annotations, and human-verified labels; developed **CreatiLayout-AM**, which uses amodal-mask supervision to improve occlusion handling [7].
- Contributed to differentiable object counting for text-to-image generation [8], cycle-consistent layout prediction and image generation [4], and quad-tree visual tokenization [6].

PLUS Lab, ShanghaiTech University

Undergraduate Researcher · Advisor: Prof. Xuming He

2022 – 2024

Shanghai, China

- Proposed an adaptive domain generalization framework for cross-modality MRI segmentation that combines a learning-free gradient-map representation with class-prior-informed test-time adaptation [10].
- Co-developed generative augmentation and uncertainty recalibration that let segmentation models generalize under domain shift while detecting semantic (OOD) shift [9].

PUBLICATIONS

* Equal contribution

- [1] **AVA-Encoder: Towards Agent-Native Video Representation Learning**
Chuyue Li, Jinpeng Yu, Haozhe Wang, Tian Xueyun, Zhijing Zhang, **Bingnan Li**, Shuqi Gu, Kan Ren, Jiaming Liu, Ruihua Huang
arXiv:2608.12313, 2026 [Paper]
- [2] **What to Edit Next: Visually Aligned Image-Editing Follow-Up Suggestions in Conversational Systems**
Zhijing Zhang, Jinpeng Yu, Xin Song, **Bingnan Li**, Chuyue Li, Changhui Du, Xiaolin Fang, Jiaming Liu, Ruihua Huang
arXiv:2608.07565, 2026 [Paper]
- [3] **Rethinking Classifier-Free Guidance in On-Policy Diffusion Distillation**
Bingnan Li, Haozhe Wang, Haozhong Xiong, Fangtai Wu, Jinpeng Yu, Yang Shi, Jiaming Liu, Ruihua Huang
arXiv:2607.24731, 2026 [Paper] [Project]
- [4] **CyCLEGen: Cycle-Consistent Layout Prediction and Image Generation in Vision Foundation Models**
Xiaojun Shan, Haoyu Shen, Yucheng Mao, Xiang Zhang, Abhay Anand, **Bingnan Li**, Haiyang Xu, Zhuowen Tu
arXiv:2603.14957, 2026 [Paper]
- [5] **DLLG: Dynamic Logit-Level Gating of LLM Experts**
Bingnan Li, Zhaoyang Zhang, Xiaozhe Liu, Yantao Shen, Shuli Jiang, Wei Xia, Zhuowen Tu, Stefano Soatto
Under review
- [6] **QuadTok: Quad-Tree Visual Tokenizer for Autoregressive Image Generation**
Yucheng Mao, Zeyuan Chen, Xiang Zhang, Divyansh Srivastava, **Bingnan Li**, Haiyang Xu, Zhuowen Tu
Under review
- [7] **OverLayBench: A Benchmark for Layout-to-Image Generation with Dense Overlaps**
Bingnan Li^{*}, Chen-Yu Wang^{*}, Haiyang Xu^{*}, Xiang Zhang, Ethan Armand, Divyansh Srivastava, Xiaojun Shan, Zeyuan Chen, Jianwen Xie, Zhuowen Tu
NeurIPS 2025, Datasets and Benchmarks Track [Paper] [Code] [Project]
- [8] **YOLO-Count: Differentiable Object Counting for Text-to-Image Generation**
Guanning Zeng, Xiang Zhang, Zirui Wang, Haiyang Xu, Zeyuan Chen, **Bingnan Li**, Zhuowen Tu
ICCV 2025, IEEE/CVF International Conference on Computer Vision [Paper] [Code]
- [9] **Generalize or Detect? Towards Robust Semantic Segmentation Under Multiple Distribution Shifts**
Zhitong Gao, **Bingnan Li**, Mathieu Salzmann, Xuming He
NeurIPS 2024, Advances in Neural Information Processing Systems [Paper] [Code]
- [10] **Gradient-Map-Guided Adaptive Domain Generalization for Cross Modality MRI Segmentation**
Bingnan Li, Zhitong Gao, Xuming He
ML4H 2023, Machine Learning for Health [Paper] [Code]

HONORS & AWARDS

- Outstanding Graduate of Shanghai (top 5%) 2024
- Outstanding Student, ShanghaiTech University (top 3%) 2021, 2023
- Merit Student, ShanghaiTech University (top 7%) 2022

SKILLS

- **Programming:** Python, C/C++, MATLAB
- **Frameworks & Tools:** PyTorch, NumPy, OpenCV, Linux, Git, multi-GPU cluster computing